stress study — highway; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]	RMSE m: mean (SD)
denser traffic / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.083 (0.012)
denser traffic / 500	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.083 (0.011)
denser traffic / 1000	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.083 (0.015)
denser traffic / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.081 (0.010)
denser traffic / 500	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.079 (0.011)
denser traffic / 1000	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.083 (0.012)
denser traffic / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.082 (0.010)
denser traffic / 500	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.083 (0.012)
denser traffic / 1000	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.085 (0.013)
denser traffic / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.027 (0.004)
denser traffic / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.026 (0.004)
denser traffic / 1000	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.026 (0.003)
weaker actuation / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.079 (0.013)
weaker actuation / 500	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.072 (0.009)
weaker actuation / 1000	LANS / 42	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.101 (0.130)
weaker actuation / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.077 (0.009)
weaker actuation / 500	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.071 (0.008)
weaker actuation / 1000	LANS / 43	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.098 (0.097)
weaker actuation / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.078 (0.011)
weaker actuation / 500	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.073 (0.011)
weaker actuation / 1000	LANS / 44	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.103 (0.129)
weaker actuation / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.027 (0.003)
weaker actuation / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.024 (0.004)
weaker actuation / 1000	MPC / actor-free	2.0 [0.4, 10.5]	98.0 [89.5, 99.6]	0.024 (0.005)

n=50 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

stress study — merge; per-seed estimates
*dense Merge uses the approved sampler amendment

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]	RMSE m: mean (SD)
denser traffic* / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.093 (0.060)
denser traffic* / 500	LANS / 42	0.0 [0.0, 7.1]	96.0 [86.5, 98.9]	0.132 (0.112)
denser traffic* / 1000	LANS / 42	0.0 [0.0, 7.1]	40.0 [27.6, 53.8]	0.430 (0.227)
denser traffic* / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.081 (0.037)
denser traffic* / 500	LANS / 43	0.0 [0.0, 7.1]	96.0 [86.5, 98.9]	0.140 (0.119)
denser traffic* / 1000	LANS / 43	0.0 [0.0, 7.1]	40.0 [27.6, 53.8]	0.425 (0.229)
denser traffic* / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.088 (0.052)
denser traffic* / 500	LANS / 44	0.0 [0.0, 7.1]	96.0 [86.5, 98.9]	0.129 (0.111)
denser traffic* / 1000	LANS / 44	0.0 [0.0, 7.1]	40.0 [27.6, 53.8]	0.426 (0.229)
denser traffic* / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.024 (0.005)
denser traffic* / 500	MPC / actor-free	0.0 [0.0, 7.1]	96.0 [86.5, 98.9]	0.087 (0.123)
denser traffic* / 1000	MPC / actor-free	0.0 [0.0, 7.1]	40.0 [27.6, 53.8]	0.390 (0.180)
weaker actuation / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.063 (0.008)
weaker actuation / 500	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.095 (0.012)
weaker actuation / 1000	LANS / 42	20.0 [11.2, 33.0]	28.0 [17.5, 41.7]	0.585 (0.486)
weaker actuation / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.065 (0.009)
weaker actuation / 500	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.095 (0.014)
weaker actuation / 1000	LANS / 43	20.0 [11.2, 33.0]	28.0 [17.5, 41.7]	0.585 (0.491)
weaker actuation / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.065 (0.009)
weaker actuation / 500	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.094 (0.014)
weaker actuation / 1000	LANS / 44	20.0 [11.2, 33.0]	28.0 [17.5, 41.7]	0.585 (0.486)
weaker actuation / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.023 (0.003)
weaker actuation / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.081 (0.028)
weaker actuation / 1000	MPC / actor-free	12.0 [5.6, 23.8]	28.0 [17.5, 41.7]	0.476 (0.269)

n=50 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

stress study — intersection; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]	RMSE m: mean (SD)
denser traffic / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.029 (0.012)
denser traffic / 500	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.041 (0.030)
denser traffic / 1000	LANS / 42	2.0 [0.4, 10.5]	98.0 [89.5, 99.6]	0.055 (0.079)
denser traffic / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.033 (0.011)
denser traffic / 500	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.043 (0.027)
denser traffic / 1000	LANS / 43	2.0 [0.4, 10.5]	98.0 [89.5, 99.6]	0.058 (0.078)
denser traffic / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.029 (0.012)
denser traffic / 500	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.041 (0.029)
denser traffic / 1000	LANS / 44	2.0 [0.4, 10.5]	98.0 [89.5, 99.6]	0.056 (0.079)
denser traffic / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.028 (0.032)
denser traffic / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.045 (0.047)
denser traffic / 1000	MPC / actor-free	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.060 (0.080)
weaker actuation / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.024 (0.007)
weaker actuation / 500	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.040 (0.049)
weaker actuation / 1000	LANS / 42	4.0 [1.1, 13.5]	96.0 [86.5, 98.9]	0.050 (0.071)
weaker actuation / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.029 (0.006)
weaker actuation / 500	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.044 (0.048)
weaker actuation / 1000	LANS / 43	4.0 [1.1, 13.5]	96.0 [86.5, 98.9]	0.054 (0.070)
weaker actuation / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.025 (0.008)
weaker actuation / 500	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.040 (0.048)
weaker actuation / 1000	LANS / 44	4.0 [1.1, 13.5]	96.0 [86.5, 98.9]	0.051 (0.071)
weaker actuation / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.016 (0.027)
weaker actuation / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.035 (0.057)
weaker actuation / 1000	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.042 (0.062)

n=50 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.

stress study — roundabout; per-seed estimates

Profile / FIFO (ms)	Controller / actor	Collision % [95% CI]	Corridor completion % [95% CI]	RMSE m: mean (SD)
denser traffic / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.091 (0.021)
denser traffic / 500	LANS / 42	0.0 [0.0, 7.1]	88.0 [76.2, 94.4]	0.337 (0.185)
denser traffic / 1000	LANS / 42	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.161 (0.380)
denser traffic / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.098 (0.020)
denser traffic / 500	LANS / 43	0.0 [0.0, 7.1]	86.0 [73.8, 93.0]	0.338 (0.185)
denser traffic / 1000	LANS / 43	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.144 (0.347)
denser traffic / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.093 (0.021)
denser traffic / 500	LANS / 44	0.0 [0.0, 7.1]	86.0 [73.8, 93.0]	0.337 (0.185)
denser traffic / 1000	LANS / 44	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.146 (0.349)
denser traffic / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.055 (0.007)
denser traffic / 500	MPC / actor-free	0.0 [0.0, 7.1]	96.0 [86.5, 98.9]	0.306 (0.087)
denser traffic / 1000	MPC / actor-free	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.129 (0.348)
weaker actuation / 0	LANS / 42	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.120 (0.009)
weaker actuation / 500	LANS / 42	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.404 (0.045)
weaker actuation / 1000	LANS / 42	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.254 (0.212)
weaker actuation / 0	LANS / 43	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.124 (0.009)
weaker actuation / 500	LANS / 43	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.408 (0.045)
weaker actuation / 1000	LANS / 43	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.254 (0.199)
weaker actuation / 0	LANS / 44	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.123 (0.011)
weaker actuation / 500	LANS / 44	0.0 [0.0, 7.1]	98.0 [89.5, 99.6]	0.406 (0.044)
weaker actuation / 1000	LANS / 44	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.253 (0.206)
weaker actuation / 0	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.100 (0.005)
weaker actuation / 500	MPC / actor-free	0.0 [0.0, 7.1]	100.0 [92.9, 100.0]	0.403 (0.038)
weaker actuation / 1000	MPC / actor-free	0.0 [0.0, 7.1]	0.0 [0.0, 7.1]	1.253 (0.202)

n=50 paired traffic cases per condition. Repeated actor seeds share traffic cases; intervals are marginal Wilson95%.
MPC is run once per condition. RMSE SD describes episode variability; paired tests are in paired_tests.csv.